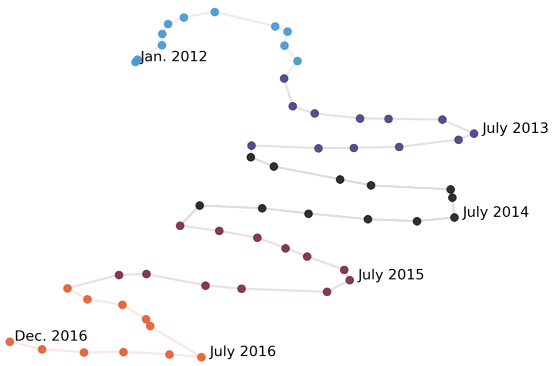

Figure 4: **Time vectors are organized in a manifold that reflects temporal variation.** Each point is a UMAP projection of the last feedforward layer of a T5-small time vector finetuned on single month of WMT. Points and edges between adjacent months are colored by year. Distances between the weights of time vectors correlate with temporal misalignment (§4.2).

veals non-linear patterns in temporal misalignment, which correspond to the cycle of months in each year. This pattern is captured by the stripes that occur parallel to the diagonal every 12 months, which indicate that the model for a particular month tends to do better on the same month in other years. We quantify these differences in perplexity in appendix Figure 12. We also report degradation patterns in *online* training settings in §A.4.

3.3 Summary

We measure temporal misalignment across a variety of domains, tasks and time scales. While performance decays linearly on a yearly scale, we discover seasonal trends in month-to-month misalignment. Next, we analyze how these phenomena relate to the weights of time-specific models, and then use that relationship to present techniques for adapting LMs to new times.

4 Temporal Adaptation with Time Vectors

The collection of year and month-finetuned models from §3 presents a new source of data to study temporal misalignment: model weights. In this section, we analyze these weights through the lens of *time vectors*, formed by taking the difference of a model finetuned on a specific time and the pretrained model. First, we show that the weights of two time vectors become less similar as the times they were finetuned on become more misaligned (§4.2). Then, we attempt to use the reverse relationship to update models to unseen times: reducing misalignment on intervening (§4.3), future (§4.4), and multiple time periods (§4.5) by interpolating time vectors.

	Pearson r		
T5 size	WMT LM	NewsSum	PoliAff
small	-0.867	0.663	0.654
large	-0.737	0.628	0.672
3b	-0.795	0.626	0.668

Table 1: **The similarity between time vectors correlates with temporal degradation.** Pearson correlation between cosine similarity of yearly time vectors and % degradation from the mean performance of all yearly models on each evaluation time period. All p-values are $< 8 \times 10^{-4}$.

4.1 Background and Definition

Task vectors (Ilharco et al., 2023) are the difference of the weights of a pretrained model from the weights of the same model after finetuning on a task. Adding and subtracting task vectors from finetuned models is a simple and effective way to improve performance on other settings, or reduce unwanted behavior without further training. Like word embeddings, if there are tasks with the analogous relationship "A is to B as C is to D," then task vectors can be used to improve performance on D with the approximation $D \approx C + (B - A)$.

Time vectors are an extension of task vectors to the time domain. Given the weights of the pretrained model, θ_{pre} and those of the model finetuned on data from only a single time period t, θ_t, a time vector $\tau_t = \theta_t - \theta_{\text{pre}}$. Like their task-based counterparts, we add back the pretrained weights at inference time and evaluate $\theta_{\text{pre}} + \tau_t$ (Ilharco et al., 2023). We call time vectors from models finetuned on individual years and months "year-vectors" and "month-vectors."

4.2 Correlation of Time Vector Similarity and Temporal Degradation

We visualize time vectors with a UMAP in Figure 4, which suggests that time vectors closer together in weight space are also closer together in time. To verify this hypothesis, we measure the cosine similarity between model weights from each pair of time vectors trained on different time periods (visualized in §A.1).

We find that this similarity metric and performance (Figure 11) decay similarly over time. Table 1 shows that the correlation between cosine similarity and relative performance change on different years is highest in WMT language modeling. Correlations are generally similar across T5 sizes, with a higher score for T5-small in the WMT LM setting than T5-large and T5-3b, and no absolute values less than 0.6.

This relationship also extends to the monthly scale. Seasonal stripes are visible in the cosine similarities between each pair of monthly WMT time vectors (visualized in Appendix Figure 9). The monthly performance degradation from the mean (Figure 3) and cosine similarity matrices (Figure 9) have a negative correlation (Pearson $r = -0.667$; $p < 10^{-16}$). We analyze cosine similarities to single-year time vectors throughout online training in Appendix §A.5.

These results indicate that time vectors are organized in way that is predictive of their performance on corresponding time periods. Next, we explore how we can use this structure to improve performance on new time periods by interpolating between time vectors.

4.3 Generalizing to Intervening Time Periods

Archiving issues or a low sampling rate can lead to gaps in datasets between the oldest and newest examples. Without data, we expect models to perform worse on these "gap" times due to temporal misalignment. In this section, we find that we can generalize better to these intervening time periods by interpolating between models finetuned on the oldest and newest times.

Method For two time vectors τ_j, τ_k, we compute their interpolation $\alpha \cdot \tau_j + (1 - \alpha) \cdot \tau_k$ with $\alpha \in [0, 1]$. In this section, we interpolate between the earliest year time vector τ_0 and latest year time vector τ_n and evaluate on times $t_0, ..., t_n$ for each $\alpha \in [0.1, 0.2, ..., 1.0]$.

Results Figure 5 shows that interpolating between start and end-year finetuned models improves performance on intervening years in both WMT LM and PoliAff tasks. Improvement is generally greatest on the exact middle years (2014 for WMT LM, 2017 for PoliAff) and decreases on years closer to start and end times. Patterns of improvement also vary depending on setting, with flatter changes in performance near $\alpha = 1.0, 0.0$ in PoliAff compared to WMT LM, and minimal improvements in NewsSum across αs compared

	Perplexity (↓)	*Rouge* (↑)	*F1* (↑)
Method	WMT LM	NewsSum	PoliAff
Start-year finetuned (τ_0)	13.92	38.56	0.6886
End-year finetuned (τ_n)	13.84	35.09	0.6967
$\frac{1}{2}(\tau_0 + \tau_n)$	13.77	38.86	0.7765
Best interpolations	13.75	40.11	0.7941
Eval-year finetuned (τ_i)	**13.65**	**42.36**	**0.8341**

Table 2: **Interpolation between start and end-year finetuned models reduces temporal misalignment on intervening years.** T5-3b average performance on each year between start and end (non-inclusive). "Best interpolations" use the best performing α values for each year.

to the difference in performance between evaluation years. Table 2 quantifies these changes, showing that interpolation closes the gap on intervening years between temporally aligned and misaligned models. Improvements are particularly large for PoliAff, nearly eight macro-F1 points just by averaging the start and end-year time vectors.

Figure 6 shows that these results extend to the monthly scale for WMT LM; we can interpolate between time vectors finetuned on January and December in a year to improve performance on the months between them. The best interpolations for each month follow an intuitive pattern, with a higher percentage of the January model leading to better performance on earlier months and vice versa.

4.4 Generalizing to the Future

The creation of labeled datasets lags behind corpera of raw text, which can be scraped automatically. As a result, language models that rely on supervision for finetuning are quickly outdated. Updating these models can be expensive, involving extra finetuning and creating labeled datasets from more recent examples. In this section, we present a new technique for updating task models finetuned on a source time period j to a target time period k with only unlabeled data from j, using task analogies (Ilharco et al., 2023).

Method Given language models with weights $\theta_j^{\text{LM}}, \theta_k^{\text{LM}}$ finetuned on unlabeled text from times j, k, and a task-specific model with weights θ_j finetuned on labeled data from time j, we perform the

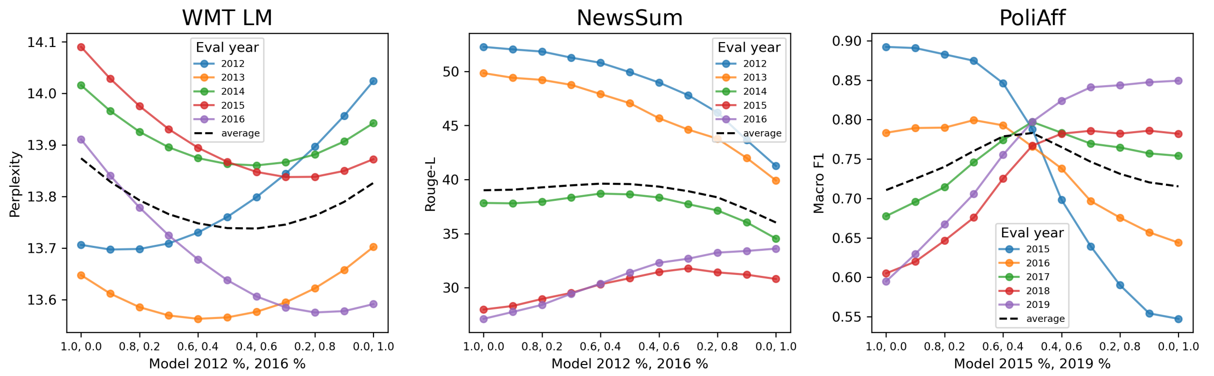

Figure 5: **Interpolating between two year vectors improves performance on the years between them.** These performance improvements follow an intuitive structure, e.g. when interpolating between 2012 and 2016, the best result on 2013 occurs with a higher percentage of 2012 and vice versa for 2015. Improvement from interpolation varies across settings.

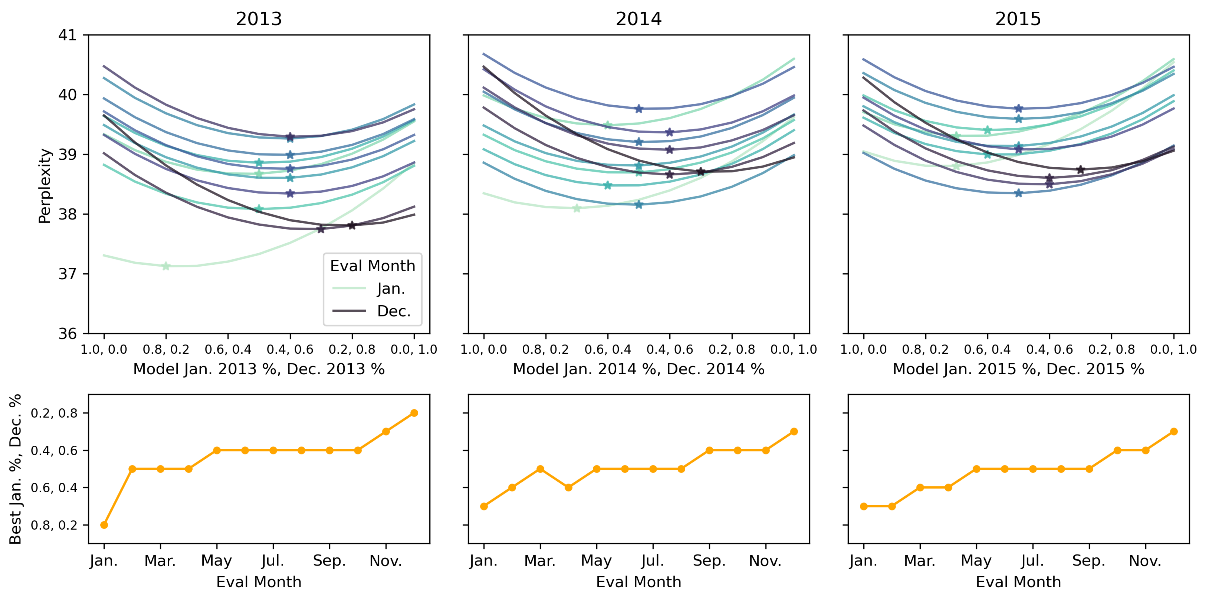

Figure 6: **Interpolating between two month vectors improves performance on the months between them.** We interpolate between January and December month vectors and evaluate on all other months within the same finetuning year. Like at the yearly scale, early months do better with a higher percentage of the January model and vice versa while middle months do best with a 50% split between the models. The stars in the upper plots correspond to the best performing interpolations for each evaluation month; these optimums are mirrored in the lower line plots.

following arithmetic on the vectors:

$$\begin{aligned}
\tau_j &= \theta_j - \theta_{pre} \\
\tau_j^{\text{LM}} &= \theta_j^{\text{LM}} - \theta_{pre} \\
\tau_k^{\text{LM}} &= \theta_k^{\text{LM}} - \theta_{pre} \\
\tau_k &\approx \alpha_1 \cdot \tau_j + (\alpha_2 \cdot \tau_k^{\text{LM}} - \alpha_3 \cdot \tau_j^{\text{LM}}) \\
\theta_k &= \tau_k + \theta_{pre}
\end{aligned}$$

We evaluate our estimated θ_k on each target time t_k, sweeping over all combinations of $\alpha_1 \in [0.6, 0.8, \ldots 2.2], \alpha_2, \alpha_3 \in [0.1, \ldots 0.6]$ and reporting the best result compared to the original model θ_j. In this section, we update a 2012 NewsSum model to 2013–2016, and a 2015 PoliAff model to 2016–2020 using WMT LM and Twitter LM time vectors respectively.

Results Task analogies improve performance on future years in both PoliAff and NewsSum tasks. Figure 7 shows that improvement compared to finetuning on the start year increases as the target and start years become more misaligned. Model size also affects performance, with T5-large and T5-3b showing greater improvements. In PoliAff, T5-